

Lab	Practical
1	Study five application similar to the project definition given and prepare competitive analysis in MS Excel. You may select a specific niche (e.g., Food Delivery or Note-taking apps)
2	Identification of project definition and analysis. Identification of project title. Write a project abstract (Short description of project minimum 150 words). Write a detailed project description (minimum 600 words). List out project requirements. • Functional requirement • Non-Functional requirement
3	Prepare a Use-Case diagram for a given project.
4	Prepare an activity diagram for a given project.
5	Prepare a sequence diagram for a given project.
6	Prepare a state diagram for a given project.
7	Prepare a class diagram for a given project.
8	DFD (Data Flow Diagram) • Prepare DFD level 0 (context) diagram for given project. • Prepare DFD level 1 diagram for a given project.
9	Project and Database Design • Prepare screen list • Prepare database schema
10	Identify test scenario for any <i>two</i> screens of the given project.
11	Write a test case based on the test scenario from practical 9.
12	Draw a control flow diagram and apply Cyclomatic Complexity. Be sure about following points • Guarantees that all independent execution paths are exercised at least once. • Guarantees that both the true and false side of all logical decisions are exercised. • Executes the loop at the boundary values and within the boundaries. • Identify numbers of independence paths required for testing. Exercise 1 <pre>void main() { int i,j,k; readln (i,j,k); if((i < j) (i > k)) { writeln("then part"); if (j < k)</pre>

	<pre> writeln ("j less than k"); else writeln ("j not less than k"); } else writeln("else Part"); } </pre> Exercise 2 <pre> i = 0; n=4; //N-Number of nodes present in the graph while (i < n-1) do j = i + 1; while (if (A[i] < A[j]) then swap(A[i], A[j]); end do; i=i+1; end do; </pre>
13	Compute Function Point Value Compute function point value for a project with the following domain characteristics: Number of user input: 5 Number of user output: 5 Number of users enquiries: 6 Number of files: 5 Number of external interfaces: 5 Assume that all the complexity adjustment values are simple. Where $\sum (Fi) = 25$ Compute Function Point Value Compute the function point for the following data: Number of user inputs = 24 Number of user outputs = 46 Number of inquiries = 8 Number of files = 4 Number of external interfaces = 2 Weighting factor are simple and where $\sum (Fi) = 35$
14	Solve using the COCOMO model estimates effort and development time. Exercise 1: Considering your immense expertise in software development, the absolute beginners Inc. has recently allotted you a mega project. - The goal of the project is to create a database of all Hindi films released since 2000. - The software would allow one to generate a list of top ten hit films, top ten flop films, best comedy films, and so on.

	- Using your prior experience you have decided the approximate sizes of each module of the software as follows: - Data entry (0.9 KDSI) - Data update (0.7 KDSI) - Query (0.9 KDSI) - Report generation and display (2 KDSI) Solve the problem by applying the basic COCOMO model - Find project type. - Find project size. - Find initial effort estimation. - Find time for development. - Find the minimum size of the team you would require to develop this system. Exercise 2: - Consider a project to develop a full screen editor. - The major components identified are - Screen edit, - Command Language Interpreter, - File input and output, - Cursor movement and - Screen movement. - The sizes for these are estimated to be 4K, 2K, 1K, 2K and 3K delivered source code lines. - Use COCOMO model to determine: Schedule estimates for different phases.
15	Introduction to Jira for Agile Project Tracking.